



Agriculture: Apple Pest Management Guidelines

Bacterial Blossom Blast

Pseudomonas syringae

Symptoms and Signs

Fruit buds infected in early bloom stop growing, turn brown and papery, and may drop off. Later infections affect flower petals and stems, and fruit cluster bases turn brown or black. Infection usually does not move beyond the base of the fruit cluster; sometimes, [fruit spurs are killed\(/PMG/P/D-AP-PSYR-BL.002.html\)](#). Flower symptoms may closely resemble fire blight, but bacterial ooze is never present with blast as it is with fire blight.

Comments on the Disease

Pseudomonas syringae is a widespread bacterium that infects many plant species and occasionally causes bacterial blossom blast on apples. Frost or freezing temperatures and free moisture are required for *Pseudomonas* infection to occur. Warm, dry weather halts its development.

Management

Bacterial blossom blast is difficult to control. When possible, use overhead sprinklers to prevent freezing. Monitor temperatures wherever freezing is most likely to occur in the orchard and turn on sprinklers when the temperature drops below 34°F (1.1°C). Turning on sprinklers before the initial drop in temperature helps avoid the damage that results from evaporative cooling. Uniform coverage of all tree parts to produce a combination of water and ice is essential. Finally, keep sprinklers going until temperatures are well above 32°F (0°C) and the danger of frost is past.

Organically Acceptable Methods

The use of overhead sprinklers to prevent freezing and sprays of approved fixed copper are organically acceptable methods.

Treatment Decisions

As an alternative to using overhead sprinklers, treat weekly with copper materials or antibiotics from green tip through bloom if blossom blast has been a problem and a freeze is likely. Apply chemicals before a freeze; treatment after symptoms appear is ineffective.

	Common name	Amount to use	REI‡	PHI‡
	(Example trade name)		(hours)	(days)
Pesticide precautions Protect water Calculate VOCs Protect bees				
Not all registered pesticides are listed. The following are ranked with the pesticides having the greatest IPM value listed first —the most effective(/agriculture/apple/Fungicide-Efficacy-for-Apple-Diseases/) and least likely to cause resistance are at the top of the table. When choosing a pesticide, consider information relating to the pesticide's properties(/agriculture/apple/General-Properties-of-Fungicides-Used-in-Apples/) and application timing(/agriculture/apple/Treatment-Timings-for-Key-Diseases/), honey bees(/bee-precaution-pesticide-ratings/), and environmental impact(/mitigation/). Always read the label of the product being used.				
A. STREPTOMYCIN SULFATE				
	(Agri-Mycin 17)	4.8 oz/100 gal	12	50
COMMENTS: Streptomycin-resistant strains of blossom blast bacteria are present in some areas. Do not exceed 48 oz/acre per application.				
B. FIXED COPPER#				
		Label rates	24	0
MODE-OF-ACTION GROUP NAME (NUMBER ¹): Multi-site contact (M1)				
COMMENTS: Application of copper-containing materials beginning at or about green tip may cause fruit russetting on some cultivars, including Granny Smith. Copper-resistant strains of blossom blast bacteria are present in some areas. Not all copper compounds are approved for use in organic production; be sure to check individual products.				

‡	Restricted entry interval (REI) is the number of hours (unless otherwise noted) from treatment until the treated area can be safely entered without protective clothing. Preharvest interval (PHI) is the number of days from treatment to harvest. In some cases the REI exceeds the PHI. The longer of two intervals is the minimum time that must elapse before harvest.
#	Acceptable for use on organically grown produce.
¹	Group numbers are assigned by the Fungicide Resistance Action Committee (FRAC)(http://www.frac.info/) according to different modes of actions. Fungicides with a different group number are suitable to alternate in a resistance management program. In California, make no more than one application of fungicides with mode-of-action Group numbers 1,4,9,11, or 17 before rotating to a fungicide with a different mode-of-action Group number; for fungicides with other Group numbers, make no more than two consecutive applications before rotating to fungicide with a different mode-of-action Group number.



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